

PYTHON FOR BEGINNERS · DAY 8

# Functions

Writing your own commands

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```
def greet(name):
    print("Hello,", name)

greet("Ali")
greet("Sara")
```

TODAY'S GOALS

# By the end of today, you can…

- Explain what a function is
- Pass in information (**parameters**)
- Get a result back with **return**
- Define one with **def** and call it
- Use default values
- Build a calculator with functions 🧮

YOU ALREADY USE FUNCTIONS!

# Python’s built-in commands

**`print()`**
shows output

**`input()`**
gets text from the user

**`len()`**
counts items

**`int()`**
converts to a number

**`sum()`**
adds up a list

**`your_own()`**
today! 🛠️

WHY FUNCTIONS?

# Write once, use many times

### Reuse

No copy-pasting the same code

### Read

A good name explains the job

### Fix

Fix a bug in one place

### Organise

Split big problems into small pieces

A function is like a **recipe** 📜: write it once, “cook” it whenever you want.

DEFINE, THEN CALL

# Your first function

**Code**

```
# 1. define it
def say_hello():
    print("Hello!")
    print("Welcome 🐍")

# 2. call it
say_hello()
say_hello()
```

**Output**

```
Hello!
Welcome 🐍
Hello!
Welcome 🐍
```

**def** + name + **()** + colon, then indent.

PARAMETERS

# Giving a function information

**Code**

```
def introduce(name, age):
    print(f"I am {name}, {age}.")

introduce("Ali", 20)
introduce("Sara", 19)
```

**Output**

```
I am Ali, 20.
I am Sara, 19.
```

**Parameters** = the names in the definition: **name, age**

**Arguments** = the values you pass in: **"Ali", 20**

DEFAULT VALUES

# A value to use if none is given

**Code**

```
def greet(name, greeting="Hello"):
    print(f"{greeting}, {name}!")

greet("Ali")
greet("Sara", "Salaam")
```

**Output**

```
Hello, Ali!
Salaam, Sara!
```

No greeting given → **"Hello"** is used.

RETURN

# Getting a result back

**Code**

```
def add(a, b):
    return a + b

result = add(3, 5)
print(result)
print(add(10, 20) * 2)
```

**Output**

```
8
60
```

The result can be **stored** and **used** in more calculations.

⚠️ COMMON CONFUSION

# print shows, return gives back

A function without **return** gives back **None** (nothing).

```
def add_print(a, b):
    print(a + b)

def add_return(a, b):
    return a + b

x = add_print(2, 3)
y = add_return(2, 3)
print(x, y)  # None 5
```

🧩 PUT IT TOGETHER

# A grade function

```
def grade(score):
    if score ⩾ 80:
        return "A"
    elif score ⩾ 50:
        return "Pass"
    return "Fail"

for s in [95, 62, 30]:
    print(s, grade(s))
```

```
95 A
62 Pass
30 Fail
```

Functions + if/elif + loops + lists — all working together.

The notebook builds a full **calculator** this way.

🧠 QUICK CHECK

# Predict before you run!

### Q1. What is printed?

```
def double(x):
  return x * 2

print(double(double(3)))
```

### Q2. Why None?

```
def square(x):
  print(x * x)

r = square(4)
print(r)
```

### Q3. What is printed?

```
def hi(n="friend"):
  return "Hi " + n

print(hi())
```

SUMMARY

# What we learned today

| Code | Meaning |
| --- | --- |
| def greet(): | define a function |
| greet() | call (run) it |
| def greet(name): | a parameter |
| def greet(name="friend"): | a default value |
| return a + b | give back a result |
| no return | the function gives back None |

💻 YOUR TURN

# Practice time!

Open **exercises.ipynb** in the Day 8 folder and solve the 10 tasks, from ⭐ to ⭐ ⭐ ⭐.

⭐ square(n) and birthday wishes

⭐ ⭐ Temperature converter

⭐ ⭐ ⭐ Report card 📄

**Next lesson:** Dictionaries & tuples 🗂️